

Similarity by Selected Traits

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Repair Bias in Data

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Missing Data

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Choose Informative Questions

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Compute distances for Distance between feature vectors. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Distance between feature vectors. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Rank independently

Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate distances, select the closest, and identify a tie among the others.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Represent books as [animal,short,illustrated] with 1=yes. Query=[1,1,0]; A=[1,0,0], B=[1,1,1], C=[0,0,0]. Count mismatches. What changes if illustrations matter twice as much?

6. Solve the given case

Traits are [animals,adventure]. A=[1,1], B=[1,0], C=[0,0]; query=[1,0].

Use number of differing traits as distance.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Score similarity using one point for same topic and one for same difficulty. Target is space, easy. A is space, hard; B is animals, easy. What happens?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 21 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Make selected traits and their scales visible in every recommendation explanation.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Represent books as [animal,short,illustrated] with 1=yes. Query=[1,1,0]; A=[1,0,0], B=[1,1,1], C=[0,0,0]. Count mismatches. What changes if illustrations matter twice as much?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Read a vector

For [4, 1] under the worked feature order, identify energy and group-work ratings.

My reasoning and evidence

13. W2 Compute distances

From target [1, 2], find distances to A [2, 2] and B [0, 4].

My reasoning and evidence

14. W3 Repair feature order

One record uses [group work, energy] while others use [energy, group work]. Why is the distance unreliable?

My reasoning and evidence

15. W4 Rank independently

Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate distances, select the closest, and identify a tie among the others.

My reasoning and evidence

16. W5 Limit similarity

Two activities have distance zero. Must they suit every learner equally well?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

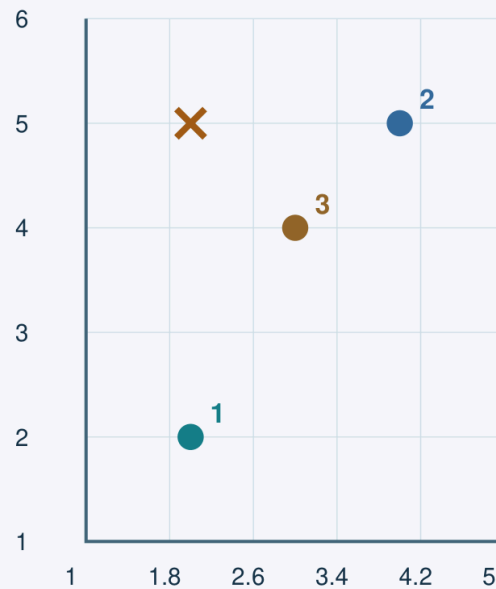
18. Pictorial case practice: Similarity by Selected Traits

A fictional toy book matcher represents two preferences with numeric features.

Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

Use Manhattan distance: add absolute coordinate differences; return all smallest-distance candidates.

Toy feature positions



Horizontal: first toy feature; vertical: second toy feature

1: A at (2, 2)

2: B at (4, 5)

3: C at (3, 4)

Cross: Target at (2, 5)

Distance rule: Manhattan distance; horizontal=first feature, vertical=second feature

Similarity by Selected Traits

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Distance A

$$|2-2|+|5-2|=0+3=3.$$



2. Distance B

$$|2-4|+|5-5|=2+0=2.$$



3. Distance C

$$|2-3|+|5-4|=1+1=2.$$



4. Choose closest

B and C tie at distance 2; keep the tie visible.

CHANGE ONE THING

Give the first-coordinate difference weight 2 while the second keeps weight 1.

Explain the changed result using the matching lesson.

Explain why B ceases to be a closest option without any candidate data changing.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches.